

Time Series — Solutions

Practice Exam — Week 9: Multivariate Time Series & VAR
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Problem 1 (5 points)

(a) The symmetry relation is

$$\Gamma_{-\tau} = \Gamma_{\tau}^{\top} \iff \gamma_{-\tau}^{(j,k)} = \gamma_{\tau}^{(k,j)}.$$

Proof. Covariance is symmetric in its two arguments, and the process is stationary, so for every j, k ,

$$\gamma_{\tau}^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)}) = \text{Cov}(X_t^{(k)}, X_{t+\tau}^{(j)}) \stackrel{s=t+\tau}{=} \text{Cov}(X_{s-\tau}^{(k)}, X_s^{(j)}) = \gamma_{-\tau}^{(k,j)}.$$

Reading this off entrywise: the (j, k) entry of Γ_{τ} equals the (k, j) entry of $\Gamma_{-\tau}$, i.e. $\Gamma_{-\tau} = \Gamma_{\tau}^{\top}$.

Why not even. Flipping the sign of τ forces a *swap of the channel indices* $(j, k) \rightarrow (k, j)$; only the diagonal entries ($j = k$) are therefore even in τ . For $j \neq k$ one has merely $\gamma_{-\tau}^{(1,2)} = \gamma_{\tau}^{(2,1)}$, which differs from $\gamma_{\tau}^{(1,2)}$ in general — this asymmetry is exactly what encodes lead-lag (directional) relationships between the two channels.

(b) Expand the cross-covariance by bilinearity, using independence of $U^{(1)}, U^{(2)}, \phi$ (all cross terms involving distinct independent zero-mean series vanish):

$$\begin{aligned} \gamma_{\tau}^{(1,2)} &= \text{Cov}(U_{t+\tau}^{(1)} + \phi_{t+\tau}, U_t^{(2)} + \phi_t) \\ &= \underbrace{\text{Cov}(U_{t+\tau}^{(1)}, U_t^{(2)})}_0 + \underbrace{\text{Cov}(U_{t+\tau}^{(1)}, \phi_t)}_0 + \underbrace{\text{Cov}(\phi_{t+\tau}, U_t^{(2)})}_0 + \text{Cov}(\phi_{t+\tau}, \phi_t). \end{aligned}$$

The first term is 0 because $U^{(1)}, U^{(2)}$ are independent, the two middle terms because ϕ is independent of the U 's, and $\text{Cov}(\phi_{t+\tau}, \phi_t) = \sigma_{\phi}^2 \delta_{\tau,0}$ because ϕ is white noise. Hence

$$\gamma_{\tau}^{(1,2)} = \sigma_{\phi}^2 \delta_{\tau,0} = \begin{cases} \sigma_{\phi}^2 & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

The marginal variances are $\gamma_0^{(1,1)} = \text{Var}(U_t^{(1)} + \phi_t) = \sigma_1^2 + \sigma_{\phi}^2$ and likewise $\gamma_0^{(2,2)} = \sigma_2^2 + \sigma_{\phi}^2$. Therefore

$$\rho_{\tau}^{(1,2)} = \frac{\gamma_{\tau}^{(1,2)}}{\sqrt{(\sigma_1^2 + \sigma_{\phi}^2)(\sigma_2^2 + \sigma_{\phi}^2)}} = \begin{cases} \frac{\sigma_{\phi}^2}{\sqrt{(\sigma_1^2 + \sigma_{\phi}^2)(\sigma_2^2 + \sigma_{\phi}^2)}} & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

The shared shock injects correlation *only at lag 0* (contemporaneously); at all nonzero lags the two channels are uncorrelated. (This is the “common shock $\Rightarrow$ everything moves together” mechanism: Γ_0 gains the rank-one term $\sigma_{\phi}^2 \mathbf{1}\mathbf{1}^{\top}$.)

(c) With $\sigma_1^2 = 3$, $\sigma_2^2 = 1$, $\sigma_{\phi}^2 = 1$: $\sigma_1^2 + \sigma_{\phi}^2 = 4$ and $\sigma_2^2 + \sigma_{\phi}^2 = 2$, so

$$\rho_0^{(1,2)} = \frac{1}{\sqrt{4 \cdot 2}} = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.3536, \quad \rho_{\tau}^{(1,2)} = 0 \quad (\tau \neq 0).$$

Problem 2 (5 points)

(a) Since $\{\varepsilon_t\}$ is uncorrelated with $\{X_t^{(2)}\}$ at all lags, $\text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)}) = 0$, so by bilinearity

$$\gamma_\tau^{(1,2)} = \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) = \alpha \text{Cov}(X_{t+\tau-\nu}^{(2)}, X_t^{(2)}) + \underbrace{\text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)})}_0 = \alpha \gamma_{\tau-\nu}^{(2,2)}.$$

The marginal variances are

$$\gamma_0^{(1,1)} = \text{Var}(\alpha X_{t-\nu}^{(2)} + \varepsilon_t) = \alpha^2 \sigma_2^2 + \sigma^2, \quad \gamma_0^{(2,2)} = \sigma_2^2,$$

(the cross term vanishes by uncorrelatedness). Hence

$$\rho_\tau^{(1,2)} = \frac{\alpha \gamma_{\tau-\nu}^{(2,2)}}{\sqrt{(\alpha^2 \sigma_2^2 + \sigma^2)} \sqrt{\sigma_2^2}} = \frac{\rho_{\tau-\nu}^{(2,2)}}{\sqrt{1 + \sigma^2/(\alpha \sigma_2)^2}},$$

using $\gamma_{\tau-\nu}^{(2,2)} = \sigma_2^2 \rho_{\tau-\nu}^{(2,2)}$. So the cross-correlation is the *autocorrelation of $X^{(2)}$ shifted by ν* and shrunk by the constant factor $(1 + \sigma^2/(\alpha \sigma_2)^2)^{-1/2} \leq 1$ coming from the added noise.

(b) Fourier-transform the cross-covariance and re-index $\tau \rightarrow \tau + \nu$ to pull out the delay phase:

$$\begin{aligned} S_{1,2}(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} = \alpha \sum_{\tau \in \mathbb{Z}} \gamma_{\tau-\nu}^{(2,2)} e^{-2\pi i f \tau} \\ &= \alpha e^{-2\pi i f \nu} \sum_{\tau' \in \mathbb{Z}} \gamma_{\tau'}^{(2,2)} e^{-2\pi i f \tau'} = \alpha \mathcal{S}_2(f) e^{-2\pi i f \nu}. \end{aligned}$$

Because $\mathcal{S}_2(f) \geq 0$ is real (it is the spectral density of $X^{(2)}$) and $\alpha > 0$,

$$\boxed{|S_{1,2}(f)| = \alpha \mathcal{S}_2(f), \quad \arg S_{1,2}(f) = -2\pi \nu f.}$$

The phase is *linear in f* precisely because a pure time shift of ν samples in the time domain becomes a multiplicative factor $e^{-2\pi i f \nu}$ in the frequency domain (the Fourier shift rule); its slope is $-2\pi \nu$, the textbook signature of a lead/lag of ν samples. The amplitude is just the spectrum of $X^{(2)}$ scaled by the gain α .

(c) With $S_{1,1}(f) = \alpha^2 \mathcal{S}_2(f) + \sigma^2$, $S_{2,2}(f) = \mathcal{S}_2(f)$ and $|S_{1,2}(f)| = \alpha \mathcal{S}_2(f)$,

$$|r_{1,2}(f)| = \frac{|S_{1,2}(f)|}{\sqrt{S_{1,1}(f) S_{2,2}(f)}} = \frac{\alpha \mathcal{S}_2(f)}{\sqrt{(\alpha^2 \mathcal{S}_2(f) + \sigma^2) \mathcal{S}_2(f)}} = \frac{\alpha \sqrt{\mathcal{S}_2(f)}}{\sqrt{\alpha^2 \mathcal{S}_2(f) + \sigma^2}}.$$

Squaring,

$$|r_{1,2}(f)|^2 = \frac{\alpha^2 \mathcal{S}_2(f)}{\alpha^2 \mathcal{S}_2(f) + \sigma^2} \in [0, 1].$$

As $\sigma^2 \rightarrow 0$ this $\rightarrow \frac{\alpha^2 \mathcal{S}_2(f)}{\alpha^2 \mathcal{S}_2(f)} = 1$ at every frequency where $\mathcal{S}_2(f) > 0$: a *pure* delay produces perfect coherence (magnitude 1) with linear phase $-2\pi \nu f$. The added noise ε_t is the only thing that pulls the coherence below 1, and it does so by the factor $(1 + \sigma^2/(\alpha^2 \mathcal{S}_2(f)))^{-1}$.

Problem 3 (5 points)

(a) Build $M(z) = I_2 - \Phi_1 z - \Phi_2 z^2$:

$$M(z) = \begin{pmatrix} 1 - 0.5z & 0 \\ -0.3z - 0.1z^2 & 1 - 0.7z + 0.1z^2 \end{pmatrix}.$$

This matrix is lower triangular, so its determinant is the product of the diagonal entries (the off-diagonal cross-terms $\Phi_1^{(2,1)}, \Phi_2^{(2,1)}$ do *not* enter the determinant):

$$\det M(z) = (1 - 0.5z)(1 - 0.7z + 0.1z^2).$$

Roots. The first factor gives $z = 2$. The quadratic $0.1z^2 - 0.7z + 1 = 0$ has

$$z = \frac{0.7 \pm \sqrt{0.49 - 0.4}}{0.2} = \frac{0.7 \pm 0.3}{0.2} \implies z = 2 \text{ or } z = 5.$$

So the three roots are $z \in \{2, 2, 5\}$, all with $|z| > 1$ (strictly outside the unit circle). Hence the VAR(2) is **stable** (and therefore stationary). *Sanity check:* $\det M(0) = \det I_2 = 1$, as required.

Companion form. Stacking $Y_t = (X_t^\top, X_{t-1}^\top)^\top \in \mathbb{R}^4$ turns the VAR(2) into the VAR(1) $Y_t = FY_{t-1} + U_t$ with

$$F = \begin{pmatrix} \Phi_1 & \Phi_2 \\ I_2 & 0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0.1 & -0.1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

The companion test is “all eigenvalues of F inside the unit circle”, and it is the reciprocal of the root test: $\det(I_2 - \Phi_1 z - \Phi_2 z^2) = z^4 \det(z^{-1}I_4 - F)$ up to a constant, so the eigenvalues of F are the reciprocals of the roots z_i . Here $\lambda = 1/2, 1/2, 1/5$ (and a structural $\lambda = 0$ from the rank-deficient Φ_2), all with $|\lambda| < 1$, giving the *same* conclusion: stable.

(b) Since $\Phi = \text{diag}(\frac{1}{2}, \frac{1}{3})$ has spectral radius $\frac{1}{2} < 1$, the VAR(1) is stable and back-substitution gives the matrix geometric (MA(∞)) filter

$$X_t = \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j}, \quad \Phi^j = \text{diag}(2^{-j}, 3^{-j}).$$

By the VAR(1) covariance formula $\Gamma_0 = \sum_{k \geq 0} \Phi^k \Sigma (\Phi^k)^\top$ with $\Sigma = I_2$ (everything diagonal, so the two channels decouple into independent AR(1)'s),

$$\Gamma_0 = \sum_{k=0}^{\infty} \text{diag}(4^{-k}, 9^{-k}) = \text{diag}\left(\frac{1}{1 - \frac{1}{4}}, \frac{1}{1 - \frac{1}{9}}\right) = \boxed{\text{diag}\left(\frac{4}{3}, \frac{9}{8}\right)}.$$

(Equivalently, solve the Lyapunov equation $\Gamma_0 = \Phi \Gamma_0 \Phi^\top + I_2$ componentwise: $g_{jj} = \phi_j^2 g_{jj} + 1 \implies g_{jj} = 1/(1 - \phi_j^2)$.) The lag-1 matrix follows from Yule–Walker $\Gamma_1 = \Phi \Gamma_0$:

$$\Gamma_1 = \text{diag}\left(\frac{1}{2}, \frac{1}{3}\right) \text{diag}\left(\frac{4}{3}, \frac{9}{8}\right) = \boxed{\text{diag}\left(\frac{2}{3}, \frac{3}{8}\right)}.$$

(c) Both Φ and $\Sigma = I_2$ are diagonal, so by the formula in (b) *every* Γ_τ is diagonal: all off-diagonal cross-covariances vanish at every lag. Hence the cross-spectrum $S_{1,2}(f) = \sum_\tau \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} \equiv 0$, and the coherence

$$r_{1,2}(f) = \frac{S_{1,2}(f)}{\sqrt{S_{1,1}(f)S_{2,2}(f)}} = 0 \quad \text{for all } f.$$

The two channels are completely uncoupled: the off-diagonal entries of Φ (here zero) are precisely the cross-channel dynamics, and there is no shared innovation (Σ diagonal), so nothing links the series at any frequency.

Problem 4 (5 points)

(a) Write $Y_t = X_t$ (zero mean). Expand the squared modulus as a double sum over T :

$$\hat{S}_h^{(p)}(f) = |J_h(f)|^2 = \left(\sum_{t \in T} h_t X_t e^{-2\pi i t f} \right) \left(\sum_{s \in T} h_s X_s e^{2\pi i s f} \right) = \sum_{t \in T} \sum_{s \in T} h_t h_s X_t X_s e^{-2\pi i f(t-s)}.$$

Take expectations, using $\mathbb{E}[X_t X_s] = \gamma_{t-s}$ and grouping the terms by the lag $\tau = t - s$ (with h set to 0 outside T , the sum over t runs over all of $\mathbb{Z}$):

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sum_{t,s} h_t h_s \gamma_{t-s} e^{-2\pi i f(t-s)} = \sum_{\tau \in \mathbb{Z}} w_\tau \gamma_\tau e^{-2\pi i f \tau}, \quad w_\tau = \sum_{t \in \mathbb{Z}} h_t h_{t+\tau}.$$

The weight w_τ is the autocorrelation of the taper. Writing $\tilde{h}_t = h_{-t}$, we have $w = \tilde{h} * h$, whose discrete Fourier transform is, by the convolution theorem,

$$W(f) = \tilde{H}(f) H(f) = \overline{H(f)} H(f) = |H(f)|^2$$

(using $\tilde{H}(f) = \overline{H(f)}$ for real h). Since $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sum_\tau w_\tau \gamma_\tau e^{-2\pi i f \tau}$ is the transform of the *product* $w_\tau \gamma_\tau$, the convolution theorem (product in time = convolution in frequency) gives

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') W(f - f') df' = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df',$$

as claimed. The kernel that blurs the true spectrum is $|H(f)|^2$.

(b) For white noise $\mathcal{S} \equiv \sigma^2$, part (a) and periodicity give

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \sigma^2 |H(f - f')|^2 df' = \sigma^2 \int_{-1/2}^{1/2} |H(f')|^2 df' = \sigma^2 \sum_{t \in T} h_t^2 = \sigma^2 \|h\|_2^2,$$

where the third equality is Parseval's relation. This holds for *every* f and every n . Therefore

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 = \mathcal{S}(f) \iff \|h\|_2^2 = \sum_{t \in T} h_t^2 = 1.$$

So the normalisation $\|h\|_2^2 = 1$ is exactly the condition that makes the tapered periodogram unbiased for a flat (white-noise) spectrum — this pins down the normalisation in the definition of a taper. (For the constant taper $h_t = 1/\sqrt{n}$ one has $\sum_t h_t^2 = n \cdot \frac{1}{n} = 1$, recovering the ordinary periodogram.)

(c) We need $\|h\|_2^2 = c^2 \sum_t g_t^2 = 1$. With $g = (1, 2, 1)$,

$$\sum_{t=1}^3 g_t^2 = 1^2 + 2^2 + 1^2 = 6 \implies c = \frac{1}{\sqrt{6}}, \quad h = \frac{1}{\sqrt{6}} (1, 2, 1).$$

(Check: $\frac{1}{6}(1+4+1) = 1$.) *Tapering vs. multitapering.* The two cure different defects of the raw periodogram. **Tapering** replaces the Fejér kernel $\mathcal{F}_n$ by the sharper, lower-side-lobe kernel $|H(f)|^2$, reducing *bias* (spectral leakage). It does *nothing* for the variance: a single tapered periodogram still has $\text{Var} \approx \mathcal{S}(f)^2$ and is inconsistent. **Multitapering** averages K tapered periodograms from orthonormal tapers, reducing the *variance* to $\approx \mathcal{S}(f)^2/K$ and restoring consistency (let $K \rightarrow \infty$ slowly), at the cost of a modest bias increase.